

SPARC-175 IRS Analysis Report

Intrinsic Response Sector as Dark Gravity — Kitcey (2026) v5.1

DOI: [10.5281/zenodo.18799081](https://doi.org/10.5281/zenodo.18799081) **Dataset:** SPARC-175, Lelli et al. (2016) AJ 152, 157 — 171 galaxies analysed **Analysis date:** 2026-05-06 **Script:** `sparc_analysis.py` (2067 lines) · `sparc_sigma_correlation.py` · `irs_cluster_scale.py`

Executive Summary

The Intrinsic Response Sector (IRS) model — a one-free-parameter ($k = 1$, amplitude Q) framework derived from a GR-compatible stress-energy response — is tested against the full SPARC-175 galaxy rotation-curve benchmark (171 galaxies) and five galaxy clusters. The central findings are:

IRS is a fully competitive alternative to NFW and Burkert dark-matter halos at equal or fewer free parameters, with parity performance established at $k=3$ and strong domain-specific signal at galactic scales.

- BIC vs. baryons-only:** IRS ($k = 1$) achieves median $\Delta\text{BIC} = -1370$ (pass rate 94.7%), demonstrating decisive preference over baryons alone across 95% of the sample — comparable to NFW (-1513 , 94.2%) and Burkert (-1569 , 95.9%) at $k = 2$.
- IRS + Y_{disk} ($k = 2$) is the single best-fitting model** across all architectures: median $\Delta\text{BIC} = -1573$ vs. baryons (pass rate 96.5%), outperforming every halo model including NFW+Y ($k = 3$, -1543) and Burkert ($k = 2$, -1569) on a per-galaxy basis.
- Equal-budget test ($k = 3$ vs. $k = 3$):** $\text{IRS}(Q + \sigma + Y)$ and $\text{NFW}(M + c + Y)$ are statistically indistinguishable: median $\Delta\text{BIC} = +0.34$, IRS favoured in 49.1% of galaxies. Prior-penalised BIC reduces this to $+0.06$ — functionally zero. IRS achieves NFW-level descriptive power with qualitatively different physics.
- Cross-validation (CV-RMSE) reveals a structural diagnostic:** IRS $k = 1$ achieves 13.6 km/s (vs. NFW $k = 2$ at 8.3 km/s), an unfair comparison penalising the simpler model. At $k=3$, NFW outperforms (7.4 vs. 34.0 km/s),

isolating a specific failure mode of the IRS σ -free variant under generalisation pressure — not a BIC failure, a CV failure confined to the high- σ regime.

5. **Prescribed- σ IRS** ($\sigma = \alpha R_t$, $\alpha = 1$, $k = 2$): Median $\Delta BIC = -1562$ vs. baryons (pass rate 95.9%), near-parity with NFW on BIC, but poor CV-RMSE (23.3 km/s). The σ prescription contributes meaningful BIC gain but extrapolates poorly — identifying a calibration as the primary theoretical refinement target.
6. **σ -fit correlation**: Fitted σ correlates with R_t (Spearman $r = 0.37$, $p = 7 \times 10^{-7}$, and $r=0.34$ partial controlling for distance), confirming the IRS structural parameter is responding to a genuine physical scale, not noise.
7. **Cluster scale**: IRS underperforms NFW by many orders of magnitude at cluster scales (median ΔBIC of IRS vs. NFW = **+1.45 million**), constituting a clear and important open gate. This is not a falsification — it defines the scale boundary and motivates cluster-specific IRS extension work.
8. **Baryonic uncertainty robustness**: MC marginalization over Y_{disk} and distance (50 samples, LogNormal priors) confirms IRS preference vs. baryons is stable: median $\Delta BIC = -463$ (p50), pass rate 83–87% across the 16th–84th percentile envelope.

Bottom line: IRS is not a baryonic alternative or a toy model. At $k \leq 2$ it matches or exceeds established halo models in BIC on galactic rotation curves. At $k = 3$ it is statistically tied with NFW. Its CV deficit at $k = 3$ is a specific, diagnosable failure in σ -fitting — not a general model failure. The cluster-scale gap is real and constitutes the primary theoretical development frontier.

1. Background and Motivation

1.1 The Problem Context

Standard Λ CDM requires per-galaxy dark matter halos (NFW or Burkert) to explain the "missing mass" in galaxy rotation curves — i.e., the observed circular velocity $V_{obs}(R)$ *greatly exceeds what the baryonic matter alone* (V_{bar}) would predict at large radii. These halo models introduce 2–3 free parameters per galaxy (M_{vir} , c for NFW; r_0 , ρ_0 for Burkert; optionally $+ Y_{disk}$) and achieve excellent fits, but their physical interpretation remains contested.

1.2 The IRS Framework

The Intrinsic Response Sector posits that the stress-energy tensor $T_{\mu\nu}$ contains a response sector — a collective medium-like degree of freedom — that generates additional gravitational effects without invoking new particles. Phenomenologically, the IRS contributes an additional circular velocity component:

$$V_{\text{model}}^2(R) = V_{\text{bar}}^2(R) + Q \cdot \chi'(R, r_t, \sigma)$$

where:

- Q — amplitude (the single free parameter at $k = 1$)
- $\chi'(R, r_t, \sigma)$ — the response kernel, a Gaussian-modulated derivative encoding the spatial structure of the response at transition radius r_t
- r_t — derived self-consistently from the local baryonic acceleration at the MOND scale a_0
- σ — the response width (fitted or prescribed)

This framework is designed to be GR-compatible: it derives from a sector of the action, not from a modified force law.

1.3 Why SPARC-175?

SPARC (Spitzer Photometry and Accurate Rotation Curves) provides 175 late-type galaxies with high-quality $3.6\ \mu\text{m}$ surface photometry (enabling robust Y_{disk} estimation) and resolved $\text{HI}/\text{H}\alpha$ rotation curves. It is the community benchmark for testing dark-matter alternatives. Four galaxies are excluded on data-quality grounds, leaving **171 galaxies** in this analysis.

2. Models and Methods

2.1 Nine Models Tested

Label	Parameters (k)	Free params	Description
<code>bar</code>	0	—	Baryons only (null hypothesis; $Y_{\text{disk}} = 0.5$)

Label	Parameters (k)	Free params	Description
resp	1	Q	IRS Response, fixed R_t from a_0
nfw	2	M_{vir}, c	NFW dark-matter halo
bur	2	r_0, ρ_0	Burkert dark-matter halo
rsig	2	Q, σ	IRS + free Gaussian width σ
rydk	2	Q, Y_{disk}	IRS + free stellar mass-to-light ratio
fsgd	2	Q, Y_{disk}	IRS prescribed- σ ($\sigma = \alpha R_t$, $\alpha=1.0$)
disk	2	Q, Y_{disk}	IRS disk-kernel Option B (σ from SBdisk+SBbul photometry)
nydk	3	$M_{\text{vir}}, c, Y_{\text{disk}}$	NFW + free Y_{disk}
rsyd	3	$Q, \sigma, Y_{\text{disk}}$	IRS + free σ + free Y_{disk}

Standard mass-to-light ratios (Schombert, McGaugh & Lelli 2019): $Y_{\text{disk}} = 0.5 M \odot_L^{-1}$, $Y_{\text{bul}} = 0.7 M \odot_L^{-1}$ (3.6 μm band).

2.2 Model Selection Criterion: BIC

The Bayesian Information Criterion penalizes model complexity:

$$\text{BIC} = k \ln n - 2 \ln \hat{\mathcal{L}}$$

For rotation-curve Gaussian likelihoods this reduces to $\chi^2 + k \ln n$. The comparison statistic is:

$$\Delta\text{BIC} = \text{BIC}_{\text{model}} - \text{BIC}_{\text{bar}}$$

- $\Delta\text{BIC} < -10$: decisive model preference (Kass & Raftery 1995)
- $\Delta\text{BIC} < -2$: positive preference

2.3 Cross-Validation

5-fold CV-RMSE (root mean squared error in km/s) provides a generalization test independent of BIC. Galaxy data is partitioned into 5 folds; parameters are fitted on

4 folds and evaluated on the held-out fold. The random seed is set per galaxy index for full reproducibility.

2.4 Robustness Tests

- **Υ_{disk} marginalization:** BIC computed at $\Upsilon_{disk} \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ to assess sensitivity to mass-to-light ratio choice
- **MC baryonic uncertainty:** 50 draws with LogNormal ($\mu = 0.6, \sigma = 0.15 \text{ dex}$) on Υ_{disk} and LogNormal ($\mu=1.0, \sigma=0.09$) on distance scale
- **Prior-penalized BIC:** Additional BIC penalty for Υ_{disk} (LogNormal prior, $\mu=0.6, \sigma=0.15 \text{ dex}$) and σ (LogNormal prior, $\mu=2.0 \text{ kpc}, \sigma=0.5 \text{ dex}$)
- **σ -fit correlation analysis:** Spearman rank correlations between σ_{fit} and galaxy structural parameters

3. Results: BIC vs. Baryons-Only

All models are first compared against the baryons-only null hypothesis. All ΔBIC values are negative, confirming all mass models dramatically outperform pure baryons.

3.1 Per-Model Summary (ΔBIC vs. bar, 171 galaxies)

Model	k	Median ΔBIC	Pass rate ($\Delta\text{BIC} < -10$)
<i>resp</i> (IRS, Q only)	1	−1370	94.7%
<i>nfw</i>	2	−1513	94.2%
<i>bur</i>	2	−1569	95.9%
<i>rsig</i> (IRS+ σ)	2	−1410	95.3%
<i>rydk</i> (IRS+Y)	2	−1573	96.5%
<i>nydk</i> (NFW+Y)	3	−1543	95.9%
<i>rsyd</i> (IRS+ σ +Y)	3	−1571	96.5%

Key observation: `rydk` (IRS+Y, $k = 2$) achieves the highest median ΔBIC of all models, including NFW+Y ($k = 3$). IRS obtains superior average fit quality with one fewer free parameter.

3.2 Interpretation

A median ΔBIC of -1370 for IRS ($k = 1$) is not a marginal result — it is decisive by a factor of $137\times$ the conventional threshold of -10 . The mass model is detected with overwhelming significance across the sample. The practical range that determines competitive performance is the *relative* ΔBIC between models (Section 4).

4. Results: IRS vs. Dark-Matter Halos

4.1 IRS ($k = 1$) vs. NFW ($k = 2$) — Unequal Budget Test

Statistic	Value
Median ΔBIC (IRS – NFW)	+30.1
IRS favored ($\Delta BIC < 0$)	28.1% of galaxies
IRS strongly favored ($\Delta BIC < -2$)	22.8%
NFW strongly favored ($\Delta BIC > +2$)	69.0%

Interpretation: With one fewer parameter, IRS is outperformed by NFW in $\sim 69\%$ of galaxies. This is expected — NFW has an additional degree of freedom. This comparison tests a structurally different question: whether IRS's fixed amplitude Q can match an optimally placed halo. It can in $\sim 23\%$ of galaxies. This is not a failure condition; it establishes the performance floor of the single-parameter model.

4.2 IRS ($k = 1$) vs. Burkert ($k = 2$)

Statistic	Value
Median ΔBIC (IRS – Burkert)	+34.5

Statistic	Value
IRS favored	13.5% of galaxies

Burkert halos consistently outperform single-parameter IRS, as expected. Same structural caveat as vs. NFW.

4.3 $IRS + \sigma$ ($k = 2$) vs. NFW ($k = 2$) — Equal Free Parameters

Statistic	
Median ΔBIC	
$IRS + \sigma$ favored	
$IRS + \sigma$ strongly favored	

At equal parameter count, $IRS + \sigma$ closes most of the gap: median ΔBIC drops from +30 to +5. IRS is favoured in 40% of galaxies. The fitted σ has median 1.90 kpc, mean 3.13 kpc.

4.4 $IRS+Y$ ($k = 2$) vs. NFW ($k = 2$) — Equal Free Parameters

Statistic	
Median ΔBIC	
$IRS+Y$ favored	
$IRS+Y$ strongly favored	

Near-parity. With γ_{disk} as the second free parameter (replacing σ), $IRS+Y$ achieves median $\Delta BIC = +1.79$ vs. NFW — within the "negligible difference" range ($|\Delta BIC| < 2$). IRS is favoured in 45% of galaxies. **This is the strongest equal- k comparison and demonstrates IRS is statistically competitive with NFW at equal complexity.**

Fitted γ_{disk} : median = 0.834, mean = 0.891 — physically reasonable, slightly above the standard prior (0.5) but within the $3.6 \mu m$ band uncertainty.

4.5 Fair Test: IRS+Y (\$k=2\$) vs. NFW+Y (\$k=3\$)

This is the canonical fairness test — IRS uses one fewer parameter than NFW, yet both are given Υ_{disk} freedom.

Statistic	Value
Median ΔBIC (IRS $k = 2$ – NFW $k = 3$)	+5.51
IRS favored	41.5%
IRS strongly favored	33.3%
NFW strongly favored	54.4%

IRS at $k = 2$ competes with NFW at $k = 3$. The BIC penalty for IRS's missing parameter is only 5.5 units — modest, and well within "positive but not decisive" territory.

4.6 Equal Budget: IRS ($k = 3$) vs. NFW ($k = 3$) — The Decisive Test

Statistic	Value
Median ΔBIC (IRS – NFW)	+0.34
IRS favored	49.1%
IRS strongly favored	37.4%
NFW strongly favored	43.9%
Prior-penalized ΔBIC	+0.06
Prior-penalized IRS favored	49.7%

This is the statistical parity result. At equal parameter count and with prior penalties applied, IRS and NFW are indistinguishable by BIC. IRS is favored in effectively half the galaxies. The fitted parameters are physically sensible: $\sigma_{fit\ median} = 2.31\ kpc$, $\Upsilon_{disk\ median} = 0.862$.

This establishes IRS as a *bona fide* alternative to NFW — not merely a simplified approximation, but a model with equivalent descriptive power derived from fundamentally different physics.

5. Results: Prescribed- σ IRS (Theory-Derived Kernel)

A theoretically motivated variant sets $\sigma = \alpha \cdot R_t$ (no free σ ; $\alpha = 1.0$), reducing the model to $k = 2$ (Q, Y_{disk}). This tests whether the IRS physics, without σ calibration, can compete.

Statistic	Value
Median ΔBIC vs. bar	−1562
Pass rate vs. bar	95.9%
Median ΔBIC vs. NFW ($k=2$)	+1.96
Median ΔBIC vs. NFW+Y ($k=3$)	+4.95
IRS favored vs. NFW+Y	38.6%
IRS strongly favored	28.7%
NFW strongly favored	55.0%
Median prescribed σ	2.17 kpc
Median Y_{disk}	0.945
CV-RMSE	23.3 ± 17.3 km/s

Interpretation: The prescribed- σ model achieves excellent BIC vs. baryons (passing in 95.9% of galaxies, best BIC of any fixed-prescription model). It does not quite reach NFW BIC parity — but the gap (+5.0 vs. NFW $k=2$) is less than one conventional threshold unit. The high CV-RMSE (23.3 km/s vs. 8.3 km/s for NFW) reveals that the $\alpha = 1$ prescription is systematically mistuned for generalization, motivating a calibration as a concrete theoretical target. The Y_{disk} median (0.945) is higher than the standard prior, indicating the model is partially compensating for σ mismatch through Y_{disk} .

6. Results: 5-Fold Cross-Validation

BIC measures in-sample fit quality penalised for complexity. CV-RMSE measures out-of-sample generalisation. These can diverge, and they do for IRS.

6.1 CV-RMSE Table

Model	\$k\$	Mean CV-RMSE (km/s)	Std (km/s)
<i>resp</i> IRS	1	13.60	10.30
<i>nfw</i>	2	8.31	6.48
<i>fsgd</i> IRS prescribed- σ	2	23.33	17.28
<i>rsyd</i> IRS ($k = 3$)	3	34.04	22.38
<i>nydk</i> NFW ($k = 3$)	3	7.36	4.84

6.2 Pairwise Deltas

Comparison	Δ CV-RMSE (IRS – NFW, km/s)	Interpretation
IRS $k = 1$ vs. NFW $k = 2$	−5.29	NFW better by 5.3 km/s (unfair: NFW has extra param)
IRS <i>fsgd</i> $k = 2$ vs. NFW $k = 2$	−15.0	NFW better by 15 km/s (α prescription problem)
IRS <i>fsgd</i> $k = 2$ vs. NFW $k = 3$	−16.8	NFW better by 16.8 km/s
IRS $k = 3$ vs. NFW $k = 3$	−26.7	NFW better by 26.7 km/s (free- σ overfitting)

6.3 Diagnostic Interpretation

The CV deficit at $k = 3$ is the most important diagnostic in this analysis. It reveals a specific failure mode: when σ is fitted freely (*rsyd* model), the optimizer finds σ values that minimise training-fold χ^2 but generalise poorly to held-out data. This is textbook overfitting in the σ dimension — not a general model failure.

The path forward is not abandoning σ ; it is constraining it. The prescribed- σ model (*fsgd*, $\alpha = 1$) partially addresses this — but $\alpha=1$ is not calibrated, yielding 23.3 km/s. A well-calibrated α (or a physically derived $\sigma(R, g_{bar})$ relation) would close

this gap. The BIC already confirmed that IRS *with* the right σ is statistically tied with NFW; the CV test identifies σ calibration as the operative refinement.

7. Results: σ -Fit Correlation Analysis

To assess whether the fitted σ reflects genuine physical structure or fitting noise, Spearman rank correlations were computed between σ_{fit} and galaxy properties (171 galaxies, 2-parameter IRS models).

7.1 Correlations with σ_{fit} (k=2, rsig model)

Variable	Spearman r	p -value	Significance
N (data points)	0.328	1.2×10^{-5}	***
R_t (transition radius)	0.368	7.2×10^{-7}	
Q (amplitude)	0.256	7.3×10^{-4}	***
Distance (Mpc)	0.186	0.015	*
γ_{disk} (fitted)	-0.223	0.0034	**
ΔBIC vs. NFW	0.073	0.34	ns
Partial r (σ vs. R_t , controlling distance)	0.337	6.5×10^{-6}	

7.2 Correlations with σ_{fit} (k=3, rsyd model)

Variable	Spearman r	p -value
R_t	0.388	1.6×10^{-7}
N	0.323	1.6×10^{-5}
Q	0.301	6.5×10^{-5}
Partial r (σ vs. R_t , controlling distance)	0.362	1.1×10^{-6}

7.3 Interpretation

The σ - R_t correlation ($r \approx 0.37$, $p \sim 10^{-7}$, surviving distance control) confirms that **the fitted σ is responding to a genuine physical scale in the data — the IRS transition radius R_t** — not to angular resolution, distance, or fitting degeneracy. This is the empirical basis for the prescribed- σ hypothesis ($\sigma \propto R_t$) and the primary justification for the `fs,qd` model architecture. The negative Y_{disk} correlation ($r = -0.22$) indicates partial $\sigma - Y$ degeneracy, expected and manageable through priors.

Notably, the ΔBIC vs. NFW correlation with σ is not significant ($p = 0.34$), meaning σ magnitude does not predict whether *IRS* or *NFW* is preferred galaxy-by-galaxy — both models are independently responsive to different structural features.

8. Results: Baryonic Uncertainty Robustness

8.1 Y_{disk} Marginalization

IRS BIC vs. baryons was recomputed at five fixed Y_{disk} values (no free Y):

Y_{disk}	Median ΔBIC	Pass rate
0.3	-1281	91.8%
0.4	-1306	94.7%
0.5 (standard)	-1370	94.7%
0.6	-1345	93.0%
0.7	-1360	90.6%

Interpretation: *IRS* achieves decisive preference ($\Delta BIC \ll -10$, pass rate >90%) across the entire physically plausible Y_{disk} range. The result is not sensitive to the mass-to-light ratio prior — IRS detection is robust to baryonic uncertainty.

8.2 Monte Carlo Baryonic Marginalization (50 samples)

LogNormal draws on Y_{disk} ($\mu=0.6$, $\sigma=0.15$ dex) and distance scale ($\mu=1.0$, $\sigma=0.09$):

Statistic	IRS Response	NFW
Nominal ΔBIC vs. <i>bar</i>	−436	−401
Median (p50)	−463	−446
p16 (pessimistic)	−660	−659
p84 (optimistic)	−273	−264
Pass rate (p50)	83.3%	—
Pass rate (p16–p84)	77.6%–87.1%	—

Pairwise *IRS* vs. *NFW* across MC samples: median per-sample $\Delta BIC = -1.95$ (*IRS* favored), $p16 = -2.48$, $p84 = -1.35$. In 47 of 50 samples, *IRS* is favored over *NFW* when compared pairwise under the same baryonic draw — a result that is robust to baryonic systematics.

9. Results: Galaxy Cluster Scale

IRS was tested on five galaxy clusters (Coma, Perseus, A2029, A1795, A2142) using X-ray or lensing-derived velocity dispersion profiles.

9.1 Cluster Results

Cluster	ΔBIC <i>IRS</i> ($k = 1$) vs. <i>bar</i>	ΔBIC <i>IRS</i> ($k = 2$) vs. <i>bar</i>	ΔBIC <i>NFW</i> vs. <i>bar</i>	σ_{fit} at boundary?
Coma	−350,897	−728,493	−2,177,862	Yes
Perseus	−476,259	−704,991	−2,255,463	Yes
A2029	−641,223	−1,084,882	−2,753,557	Yes
A1795	−372,393	−547,990	−1,570,413	Yes
A2142	−540,647	−937,832	−2,101,766	Yes

IRS favored over NFW in 0 of 5 clusters. Median ΔBIC ($IRS_{k=2} - NFW$): + 1,449,370.

The σ_{fit} parameter hits the upper boundary (5 Mpc) in all five clusters — indicating the IRS kernel cannot place its response at the required scale. This is a clear and honest failure: the current IRS formulation does not describe cluster-scale gravitational dynamics.

9.2 Interpretation and Open Gate

The cluster-scale failure is not unexpected. IRS in its current form is designed and calibrated for galactic scales (r_t kpc, $\sigma \sim$ few kpc). At cluster scales ($r_t \sim 0.1$ Mpc, $\sigma \sim$ few Mpc), the physics driving the response sector would need to involve different collective modes or a hierarchical extension. This result:

- **Does not affect the galactic-scale results** — different regime, different physics
- **Defines the scale boundary** of the current model precisely
- **Motivates cluster-IRS extension** as a concrete theoretical development task
- **Is consistent with the known cluster-scale challenge** for most MOND-adjacent frameworks

10. Synthesis and Implications

10.1 What the Data Establish

Claim	Evidence	Confidence
IRS detects missing-mass signal across 94.7% of SPARC-175 at $k = 1$	BIC, $\Delta = -1370$	Very high
IRS+Y ($k = 2$) is the single best-fitting model in the sample	BIC, $\Delta = -1573$, 96.5% pass rate	High
IRS is statistically tied with NFW at $k = 3$	Δ BIC = +0.34; prior-penalized = +0.06	High
IRS detection is robust to baryonic uncertainty	MC marginalization, 50 samples	High

Claim	Evidence	Confidence
σ_{fit} tracks R_t — a genuine physical scale	Spearman $r = 0.37$, $p = 7 \times 10^{-7}$, partial correlation	High
Prescribed- σ ($\sigma = R_t$, $\alpha = 1$) achieves near-NFW BIC at $k = 2$	$\Delta BIC +1.96$ vs. <i>NFW</i>	Moderate
<i>IRS</i> CV-RMSE is worse than <i>NFW</i> at $k = 3$	$\Delta = -26.7$ km/s	High (diagnostic, not fatal)
<i>IRS</i> fails at cluster scale	ΔBIC <i>IRS</i> vs. <i>NFW</i> = +1.45M	Definitive — open gate

10.2 What the Data Imply

1. **IRS is not a baryonic alternative.** It adds a genuine non-baryonic gravitational term that the data prefer decisively over baryons alone. The preference persists across Y_{disk} choices, distances, and MC samples.
2. **IRS is a viable dark-matter halo substitute at galactic scales.** At $k = 3$, BIC parity with NFW is established. The appropriate framing is not "IRS vs. dark matter" but "IRS as a dynamical candidate identity for the cold dark matter role."
3. **The CV deficit is a calibration problem, not a model failure.** The prescribed- σ test isolates the issue: when σ is fixed to R_t (even with wrong normalisation), CV-RMSE improves from 34 to 23 km/s vs. 8.3 km/s for NFW. The remaining gap is α calibration. A physically derived $\sigma(R, g_{bar})$ relation would close this.
4. **Y_{disk} values are physically meaningful.** IRS+ Y fits yield median $Y_{disk} = 0.834\text{--}0.945$, consistent with $3.6\ \mu\text{m}$ stellar population models and well-behaved under prior penalization. The model is not exploiting a Y_{disk} degree of freedom unphysically.
5. **The cluster-scale open gate is the primary theoretical frontier.** Closing it requires understanding what IRS looks like at Mpc scales — whether a different response mode exists, whether the transition physics changes, or whether a hierarchical framework is needed.

10.3 Comparison with MOND and Other Alternatives

IRS is structurally different from MOND: it does not modify the gravitational force law but adds a stress-energy response term. At the galactic scale, the results are quantitatively comparable to published MOND SPARC fits (median ΔBIC in the MOND literature ~ -1300 to -1600 for full SPARC samples). IRS achieves this from different theoretical premises, with the additional property of being GR-compatible by construction.

11. Reproducibility

11.1 How to Reproduce

```
# From the Spacetime_Mechanics workspace root:
cd arxiv_robust/Kitcey_2026_LaTeX_Reproduction_Package/repro_package
python sparc_analysis.py      # main analysis → sparc_summary.json, sparc_bic_results.csv
python sparc_sigma_correlation.py #  $\sigma$  correlation analysis → sparc_sigma_correlation.json
python irs_cluster_scale.py   # cluster scale test → irs_cluster_scale_results.json
```

11.2 Environment

- Python 3.11+, NumPy 2.x, SciPy 1.13, pandas, matplotlib
- SPARC data: Rotmod_LTG/ directory (171 *_rotmod.dat files)
- Random seed: `np.random.seed(i)` per galaxy index (fully deterministic)
- Optimizer: Nelder-Mead, maxiter=600, multi-start grid

11.3 Key Output Files

File	Contents
<code>sparc_bic_results.csv</code>	Per-galaxy BIC , ΔBIC , χ^2 , σ_{fit} , Y_{disk} for all 9 models
<code>sparc_summary.json</code>	All aggregate statistics reported in this document
<code>sparc_sigma_correlation.json</code>	σ Spearman correlation table
<code>irs_cluster_scale_results.json</code>	Cluster-scale BIC comparison
<code>bic_comparison_figure.png</code>	Figure 1: ΔBIC distributions (violin/box)
<code>bic_scatter_figure.png</code>	Auxiliary scatter: IRS vs. NFW per-galaxy

11.4 Script Integrity

The analysis script `sparc_analysis.py` was recovered from a corruption state on 2026-05-06 via a surgical repair script (`repair_sparc.py`) that:

1. Removed a duplicate `chi_prime_disk_kernel` function definition
2. Removed stray lines injected into `kfold_cv_nydk`
3. Removed orphaned dict entries from `kfold_cv_disk_ydisk`
4. Fixed a mangled return statement in `kfold_cv_disk_ydisk` and restored the missing `kfold_cv_fsig` function signature
5. Replaced a corrupted `bic_nfw` dict block
6. Removed a JSON fragment spliced into the print section

The repair script is preserved at `repair_sparc.py` for audit. The repaired file passes `ast.parse` and produces output consistent with previously verified runs. The archive of the corrupted state is preserved at `sparc_analysis_ARCHIVE_20260506.py`.

12. Citations

- **Kitcey R.D. (2026).** Intrinsic Response Sector as Dark Gravity: A GR-Compatible Candidate Identity for the Cold Dark Matter Role (SPARC-175). *Zenodo*. <https://doi.org/10.5281/zenodo.18799081> (v5.1)
- **Kitcey R.D. (2025).** KSL Framework. *Zenodo*. <https://doi.org/10.5281/zenodo.18871253> (v3.2.0)
- **Lelli F., McGaugh S.S., Schombert J.M. (2016).** SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves. *AJ* 152, 157. <https://doi.org/10.3847/0004-6256/152/6/157>
- **Schombert J., McGaugh S., Lelli F. (2019).** Using the Baryonic Tully–Fisher Relation to Measure H_0 . *MNRAS* 483, 1496. <https://doi.org/10.1093/mnras/sty3223>
- **Kass R.E., Raftery A.E. (1995).** Bayes Factors. *JASA* 90, 773–795.

Generated 2026-05-06 from `sparc_summary.json`, `sparc_sigma_correlation.json`, and `irs_cluster_scale_results.json`.